

Assignment 13

Submission by: **June 2nd - 4pm**

1) [0.6 points]

Consider the following Turing machine M that recognises odd strings over $\Sigma =$ $\{0,1\}$, $Q = \{q_0, q_1, q_{acc}, q_{rej}\}$, $\Gamma = \{0,1, \perp\}$, and the transition function:

$$\delta(q_0, 0) = \{q_0, 0, R\} \quad \delta(q_1, 0) = \{q_1, 0, R\}$$

$$\delta(q_0, 1) = \{q_1, 1, R\} \quad \delta(q_1, 1) = \{q_0, 1, R\}$$

$$\delta(q_0, \perp) = \{q_{rej}, \perp, R\} \quad \delta(q_1, \perp) = \{q_{acc}, \perp, R\}$$

Do the following:

- Construct the tableau of size 5 X 5 representing the only possible branch computation of M for the string $w = 111$
- Write down the Boolean formulas ϕ_{start} and ϕ_{accept} of the polynomial time reduction from the language of odd strings over $\Sigma = \{0,1\}$ and SAT.

2) [0.4 points]

Say which of the following windows are legal, hence may appear in a tableau representing a branch of computation of a nondeterministic Turing machine:

a)

b	q_1	b
q_2	b	q_2

b).

a	b	a
a	a	a

c)

a	q_1	b
q_2	a	c

d)

a	a	q_1
a	a	b

e)

a	q_1	b
a	a	q_2

b	b	b
c	b	b

a	b	a
a	b	q_2

f)

g)

3) [0.6 points]

In the proof of Cook-Levin Theorem, a window is a 2 X 3 rectangle of cells. Show why the proof would have failed if we had used a 2 X 2 rectangle instead.

4) [1.5 point]

Let $UHAMPATH = \{ \langle G, S, t \rangle \mid G \text{ is an undirected graph with a Hamiltonian path from } s \text{ to } t \}$.

Show that $UHAMPATH$ is NP-Complete using a reduction from $HAMPATH$.

HINT: Except for s and t you will want to replace each node u of the original directed graph G with 3 nodes in G' : u^{in}, u^{mid}, u^{out}

5) [1.5 point]

$VERTEX COVER = \{ \langle G, k \rangle \mid G \text{ is an undirected graph that has a } k - \text{node vertex cover} \}$. (A *vertex cover* is a subset of nodes such that every edge of the graph G touches one of the nodes).

Prove that the language is NP-Complete using a reduction from 3SAT.

HINT: You will need two kinds of gadget:

- a) Variable gadget: for each variable x_i you want that either it's in the cover ($x_i = \text{TRUE}$) or it isn't ($\bar{x}_i = \text{TRUE}$) – create a node for each and connect them so one of them must appear in the produced cover
- b) Clause gadget: you need to make sure that at least one variable in each clause is true – create a 3-clique with the variables of each clause and connect them to the corresponding variable gadget nodes
- c) With a correct construction you should end up with a cover of size $k = v + 2c$ where v is the number of variables in the 3SAT formula and c is the number of clauses